

is somewhat limited with ZigBee owing to the absence of interoperability among ZigBee devices. Also, the devices are notoriously hard to communicate with devices from other manufacturers. This makes ZigBee a less than ideal choice for novices in the home automation arena. If you use devices from a sole manufacturer, it shouldn't be a problem. But that may not always be a feasible option. Compounding the problem is the fact that there exist multiple versions of ZigBee that don't necessarily communicate flawlessly amongst themselves.

One of the similarities that ZigBee shares with Z-Wave is that it's low-cost technology consumes very little power. This means that the battery-operated ZigBee devices will have a long life. Since it runs on the 802.15.4



Interoperability might be an issue

wireless communication standard, ZigBee makes use of a mesh network structure which provides not just great range but also rapid communication between the ZigBee devices. Even though the latest version i.e. ZigBee 3.0 assures better interoperability (between both devices and versions), how much this manifests remains to be seen.

Wi-Fi

Perhaps the greatest advantage of Wi-Fi, which is practically ubiquitous, is that it provides a high bandwidth. This in turn gives it another advantage – this one particularly significant in home automation: plenty of manufacturers create smart home devices that function with it. Already, wireless routers are to be found in many homes – these routers work on the Wi-Fi protocol. This means that a central hub is already in place in many homes all you must do is connect compatible devices to it.



Ubiquitous but power-guzzler

The major disadvantage with this protocol is the interference and the bandwidth issues that may arise with multiple devices. If the house is filled with Wi-Fi connected devices, then these smart devices will compete for the bandwidth which in turn slows down the response. Another potential problem is that Wi-Fi takes up substan-